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EPCO-15. DETECTING HISTOLOGIC & CLINICAL GLIOBLASTOMA PATTERNS OF PROGNOSTIC RELEVANCE

Bhakti Baheti¹, Sunny Rai¹, Shubham Innani¹, Garv Mehdiratta¹, Sharath Chandra Guntuku¹, MacLean P Nasrallah², and Spyridon Bakas²;

¹University of Pennsylvania, Philadelphia, PA, USA, ²University of Pennsylvania, USA, Philadelphia, USA

BACKGROUND: Glioblastoma is the most common adult primary brain tumor. Limited improvement in patient prospects over the last 20 years reflects our limited disease understanding. Robust prognostic patient stratification from whole slide images (WSI) & clinical data using interpretable computational methods could improve disease understanding and patient management. **MATERIAL AND METHODS:** The TCGA-GBM and TCGA-LGG data collections were reclassified according to the 2021 WHO classification criteria, identifying 188 glioblastoma (IDH-wt, Gr.4) cases and stratifying them to short- (< 9 months) and long-survivors (> 13 months). H&E-stained WSI were comprehensively curated to remove artifactual content. An interpretability mechanism based on weakly-supervised attention-based multiple-instance-learning was employed. The prognostic relevance of associated clinical data was also evaluated both independently and integratively, using XGBoost and 'SHapely Additive exPlanations'. Late fusion amalgamates the advantage of WSI patterns with clinical features by averaging the output probabilities of two distinct models at the decision level. Quantitative performance evaluation was based on a 10-fold monte carlo cross-validation, with training (80%), validation (10%), and test (10%) sets. **RESULTS:** Our imaging, clinical, and integrative multi-modal models revealed AUC_{Test} of 0.68/0.72/0.75 respectively. The imaging analysis revealed correctly predicted long-survivors associated with histologically malignant areas, including necrosis, hypercellularity, atypia, infiltration, and proliferation. Correctly classified short survivors highlighted influence by aggressive infiltrative features, including leptomeningeal involvement. The clinical analysis revealed age, mutation count, and TERT promoter status as the top three features contributing to stratification, in the XGBoost model. **CONCLUSION:** Our findings support data-driven histology & data interpretation of algorithmic decisions contribute to further understanding of glioblastoma by identifying morphology patterns & clinical patterns of prognostic relevance. Integration of multiple modalities can be a powerful approach for improving the performance of AI models.